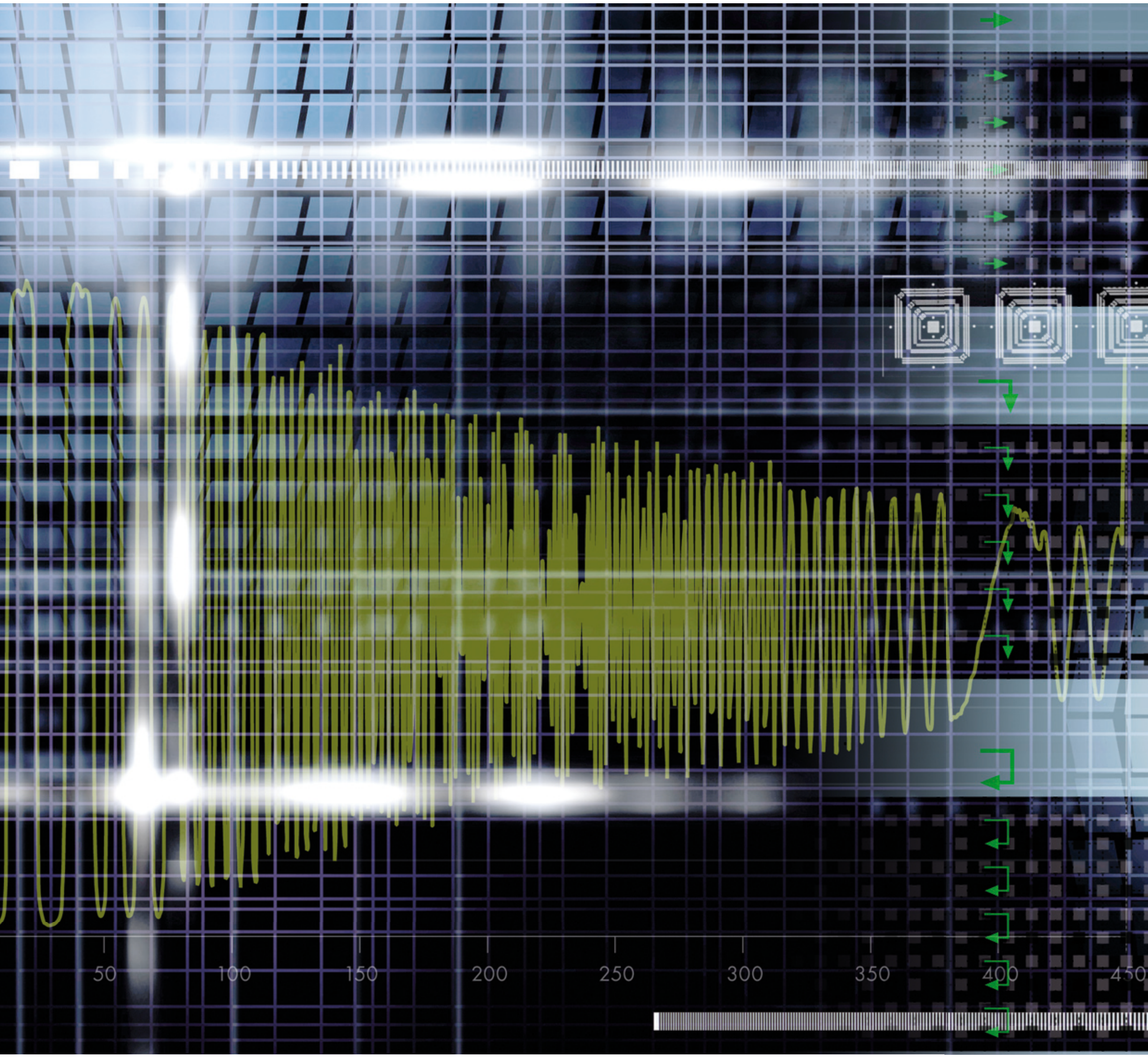
4K+ Systems

Theory Basics for Motion Picture Imaging



The Author:

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Theory Basics for Motion Picture Imaging

By Dr. Hans Kiening

INTRODUCTION

Five years ago digital cameras and cellphones were the biggest thing in consumer electronics, but the most popular phrase today seems to be “HD.” One would think that the “H” in HD stands for “Hype.”

Although professionals have a clear definition of HD (1920 x 1080 pixels, at least 24 full frames per/sec, preferably 4:4:4 color resolution), at electronics stores, the “HD-Ready” sticker on hardware, offers considerably less performance (1280 x 1024 pixels, 4:2:1, strong compression, etc.).

On display are large monitors showing images that are usually interpolated standard-definition (720 x 576 pixels or less). What the catchword HD means for the living room is, on a different quality level, comparable to the discussion of 2K versus 4K in professional postproduction, in that expectations, and sometimes just basic information could not be more contradictory. Yet an increasing number of major film projects are already being successfully completed in 4K.

So, it’s high time for a technically-based appraisal of what we really mean by 2K and 4K. This article strives to make a start.

"Seeing is believing", and of course we can‘t actually show these images on paper. To escape the limitations posed by the print environment, these sample images in their original resolution can be downloaded from our FTP server at:

Host:	**ftp2.arri.de**
login:	**4film**
Password:	**ARRI**

PART I: RESOLUTION AND SHARPNESS

To determine resolution, a raster is normally used, employing increasingly fine bars and gaps. A common example in real images would be a picket fence displayed to perspective.

In the image of the fence, shown in *Fig. 1*, it is evident that the gaps between the boards become increasingly difficult to discriminate as the distance becomes greater. This effect is the basic problem of every optical image. In the foreground of the image, where the boards and gaps haven't yet been squeezed together by the perspective, a large difference in brightness is recognized. The more the boards and gaps are squeezed together in the distance, the less difference is seen in the brightness.

Figure 1: "Real world" example for a resolution test pattern.

To better understand this effect, the brightness values are shown along the yellow arrow in an x/y diagram *(Fig. 2)*. The brightness difference seen in the y-axis is called contrast. The curve itself functions like a harmonic oscillation; because the brightness does not change over time but spatially from left to right, the x-axis is called spatial frequency.

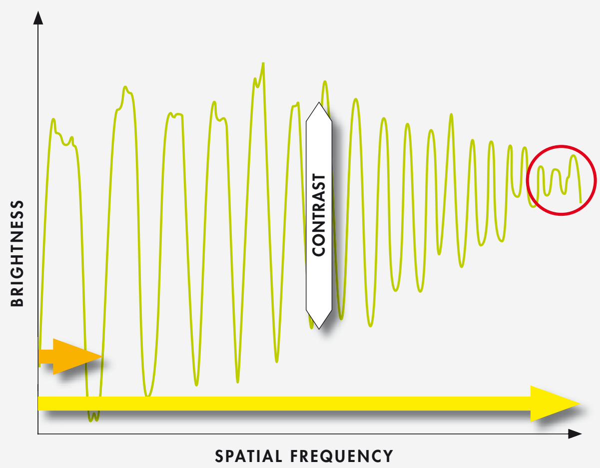


Figure 2: Brightness along the yellow arrow in Fig. 1.

The distance can be measured from board to board (orange arrow) on an exposed image such as a 35 mm film frame. It describes exactly one period in the brightness diagram (Fig. 2). If such a period in the film image continues, for example, over 0.1 mm, then we have a spatial frequency of 10 line pairs/mm (10 lp/mm, 10 cycles/mm or 10 periods/mm). Visually expressed, a line pair always consists of a bar and a "gap."

It can be clearly seen in *Fig. 1* that the finer the reproduced structure, the more the contrast will be "slurred" at that point in the image. The limit of the resolution has been reached when one can no longer clearly differentiate between the structures. This means the resolution limit (red circle indicated in *Fig. 2*) lies at the spatial frequency where there is just enough contrast left to clearly differentiate between board and gap.

Constructing the Test

Using picket fences as an example *(Fig. 1)*, the resolution can be described only in one direction. Internationally and scientifically, a system of standardized test images and line pair rasters to determine and analyze resolution has been agreed upon. Horizontal and vertical rasters are thereby distributed over the image surface.

To carry out such a test with a film camera, the setup displayed in *Fig. 4* was used. The transparent test pattern was exposed at 25 frames/sec and developed.
Fig. 5 shows the view through a microscope of the image center (orange border in *Fig. 3*)

Resolution Limit 35 mm

Camera: . . . . . ARRICAM ST
Film: . . . . Eastman EXR 50D Color Negative Film 5245 EI 50
Lens: . . . . HS 85 mm F2.8
Distance: . . . . . 1.65 m

If the image is downloaded at 35_micro.tif and viewed on a monitor with a zoom factor of 100%, it is apparent that the finest spatial frequency that can still be differentiated lies between 80 and 100 lp/mm.
For these calculations, a limit of 80 lp/mm can be assumed. The smallest discernible difference was determined by the following:

1 mm/80 lp = 0.012 mm/lp
Lines and Gaps are equally wide, ergo:
0.012 mm/2 = 0.006 mm for the smallest detail

Resolution Limit 16 mm

Camera: 416
Film: . . . . Eastman EXR 50D Color Negative Film 5245 EI 50
Lens: . . . . HS 85 mm F2.8
Distance: . . . . . 1.65 m

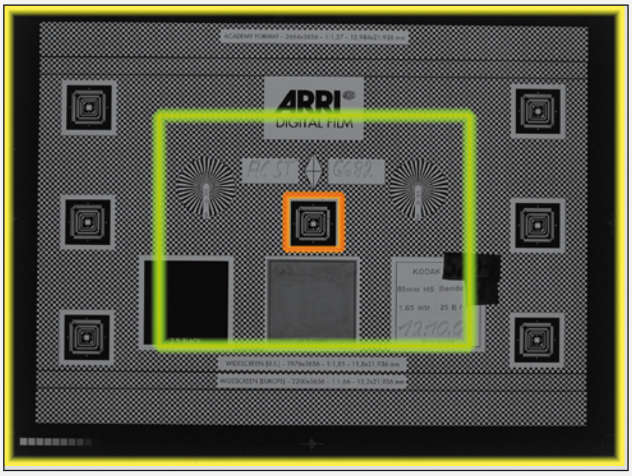


Figure 3: Area captured with 35 and 16 mm negative (yellow/green border) and cutout viewed in the microscope (orange border).

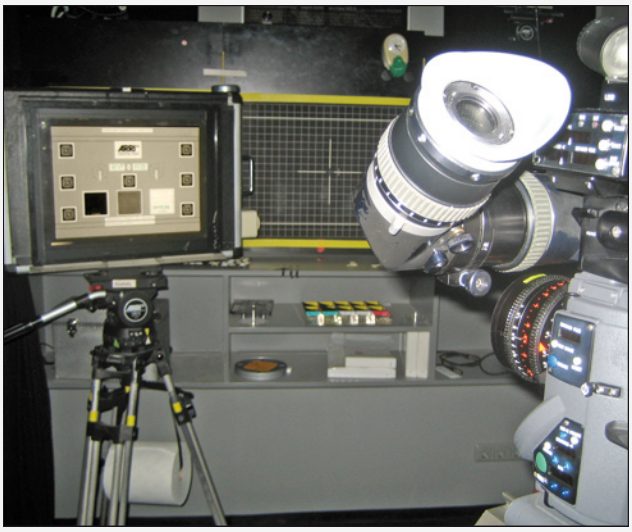


Figure 4: Setup for camera resolution test with transparent test pattern.

If a 35 mm camera is substituted for a 16 mm camera with all the parameters remaining the same (distance, lens), an image will result (image 16_micro.tif) that is only half the size of the 35 mm test image, but resolves exactly the same details in the negative.

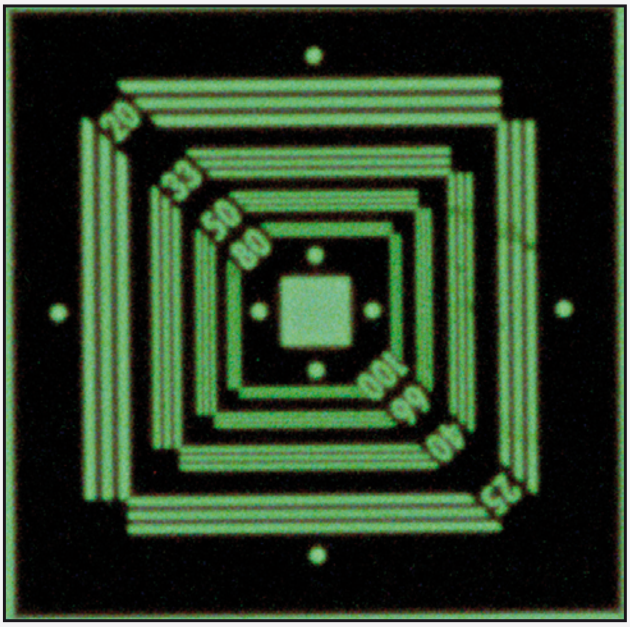


Figure 5a: Microscope view of 35 mm negative

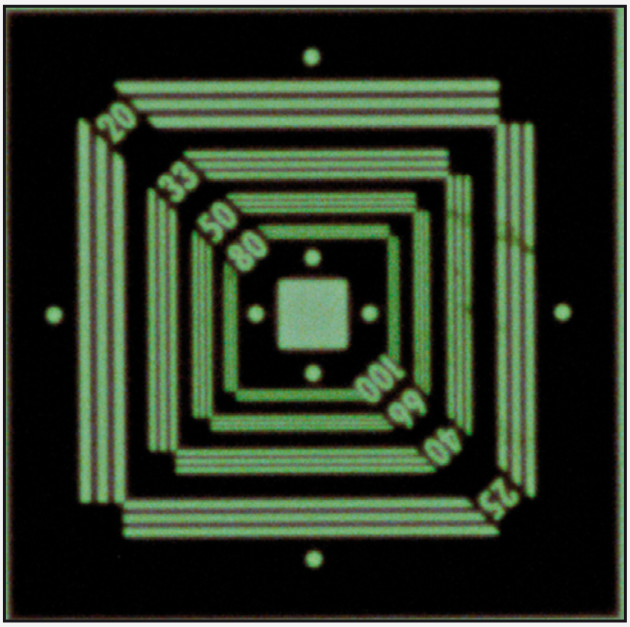


Figure 5b: Microscope view of 16 mm negative

Results

This test is admittedly an ideal case, but the ideal is the goal when testing the limits of image storage in film. In the test, the smallest resolvable detail is 0.006 mm large on the film, whether 35 mm or 16 mm. Thus, across the full film width there are 24.576 mm / 0.006 = 4096 details or points for 35 mm film and 12.35 mm / 0.006 = 2048 points for 16 mm film. These are referred to as points and not pixels because we are still operating in the analog world. These statements depend upon the following:

(1) looking at the center of the image
(2) the film sensitivity is not over 250 ASA
(3) exposure and development are correct
(4) focus is correct
(5) lens and film don't move against one another during exposure
(6) speed <50 frames/sec

Digital

The same preconditions would also exist for digital cameras (if there were a true 4K resolution camera on the market today); only the negative processing would be omitted. Thus in principle, this test is also suitable for evaluating digital cameras. In that case, however, the test rasters should flow not only horizontally and vertically, but also diagonally, and, ideally, circularly. The pixel alignment on the digital camera sensor (Bayer pattern) is rectangular in rows and columns. This allows good reproduction of details which lie in the exact same direction, but not of diagonal structures, or other details that deviate from the vertical or horizontal. This plays no role in film, because the "sensor elements"—film grain—are distributed randomly and react equally well or badly in all directions.

Figure 6: Resolution = Sharpness?

Sharpness

Are resolution and sharpness the same? By looking at the images shown in *Fig. 6*, one can quickly determine which image is sharper. Although the image on the left comprises twice as many pixels, the image on the right, whose contrast at coarse details is increased with a filter, looks at first glance to be distinctly sharper.

The resolution limit describes how much information makes up each image, but not how a person evaluates this information. Fine details such as the picket fence in the distance are irrelevant to a person's perception of sharpness—a statement that can be easily be misunderstood. The human eye, in fact, is able to resolve extremely fine details. This ability is also valid for objects at a greater distance. The decisive physiological point, however, is that fine details do not contribute to the subjective perception of sharpness. Therefore, it's important to clearly separate the two terms, resolution and sharpness.

The coarse, contour-defining details of an image are most important in determining the perception of sharpness. The sharpness of an image is evaluated when the coarse details are shown in high contrast.

A plausible reason for this can be found in evolution theory: "A monkey who jumped around in the tops of trees, but who had no conception of distance and strength of a branch, was a dead monkey, and for this reason couldn't have been one of our ancestors," said the palaeontologist and zoologist George Gaylord Simpson[1]. It wasn't the small, fine branches that were important to survival, but rather the branch that was strong enough to support our ancestors.

MTF

Modulation transfer function describes the relationship between resolution and sharpness, and is the basis for a scientific confirmation of the phenomenon described earlier. The modulation component in MTF means approximately the same as contrast. If we evaluate the contrast (modulation) not only where the resolution reaches its limit, but over as many spatial frequencies as possible and connect these points with a curve, we arrive at the so-called MTF *(Fig. 7)*.

In *Fig. 7*, the x-axis illustrates the already-established spatial frequency expressed in lp/mm on the y-axis, instead of the brightness seen in modulation. A modulation of 1 (or 100%) is the ratio of the brightness of a completely white image to the brightness of a completely black image. The higher the spatial frequency—in other words the finer the structures in the image - the lower the transferred modulation. The curve seen here shows the MTF of the film image seen in *Fig. 5* (35 mm). The film's resolution limit of 80 lp/mm (detail size 0.006 mm) has a modulation of approximately 20%.

In the 1970s, Erich Heynacher from Zeiss provided the decisive proof that humans attach more value to coarse, contour-defining details than to fine details when evaluating an image.

He found that the area below the MTF curve corresponds to the impression of sharpness perceived by the human eye (the so-called Heynacher Integral)[2]. Expressed simply, the larger the area, the higher is the perception of sharpness. It is easy to see that the coarse spatial frequencies make up the largest area of the MTF. The further right we move into the image's finer details, the smaller the area of the MTF. If we look at the camera example in *Fig. 6*, it is obvious that the red MTF curve shown in *Fig. 8* frames a larger area than the blue MTF curve, even if it shows twice the resolution.

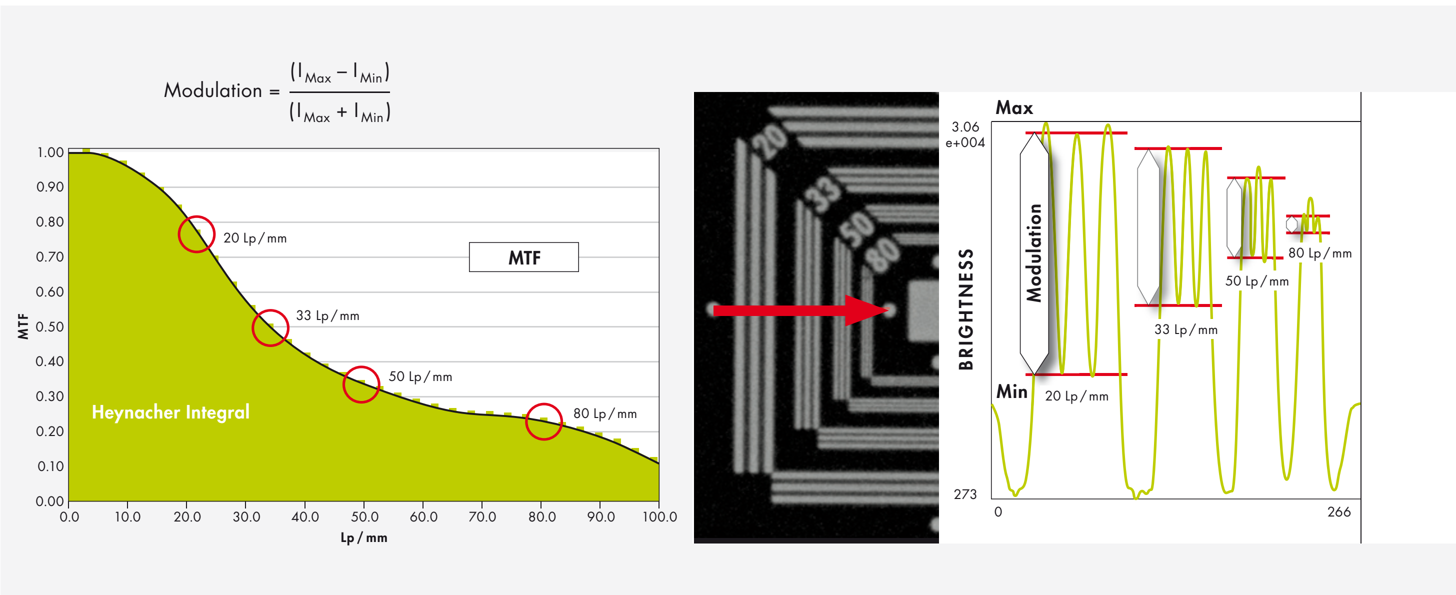


Figure 7: Modulation transfer function and the Heynacher Integral.

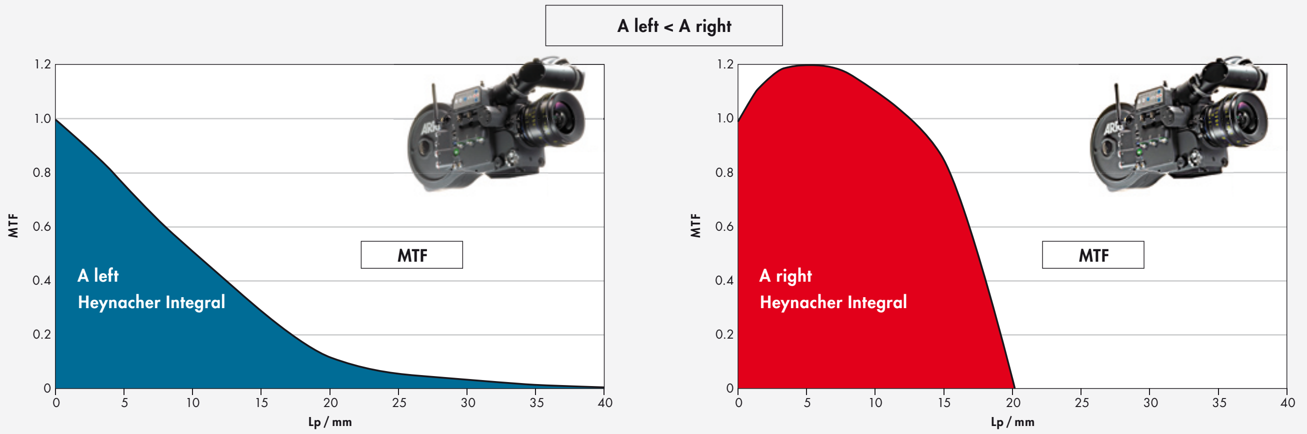


Figure 8: Heynacher Integral for both cameras in Fig. 6.

For Experts

We will forgo an explanation of the difference between sine-shaped (MTF) and rectangular (CTF) brightness distribution in such test patterns. However, all relevant MTF curves have been measured according to ISO standards Fourier Transform (FFT of slanted edges).

Summary Part I

Sharpness does not depend only on resolution. The modulation at lower spatial frequencies is essential. In other words, contrast in coarse details is significantly more important for the impression of sharpness than contrast at the resolution limit. The resolution that delivers sufficient modulation (20%) in 16 mm and 35 mm film is reached at a detail size of 0.006 mm, which corresponds to a spatial frequency of 80 lp/mm (not the resolution limit <10%).

PART II: INTO THE DIGITAL REALM

How large is the storage capacity of the film negative? How many pixels does one need to transfer this spatial information as completely as possible into the digital realm, and what are the prerequisites for a 4K chain of production? These are some of the questions that this section addreses.

The analyses here are deliberately limited to the criteria of resolution, sharpness and local information content. These, of course, are not the only parameters that determine image quality, but the notion of 4K is usually associated with them.

16 mm, 35 mm, 65 mm—
Film as a Carrier of Information

Film material always possesses the same performance data: the smallest reproducible detail (20% modulation) on a camera film negative (up to EI 200) is about 0.006 mm — as was determined by the analysis done earlier in this article. This can also be considered as the size of film's "pixels", a concept that is well known from electronic image processing. It does not matter if the film is 16 mm, 35 mm or 65 mm; the crystalline structure of the emulsion is independent of the film format. Also, the transmission capability of the imaging lens is generally high enough to transfer this spatial frequency (0.006 mm = 80 lp/mm) almost equally well for all film formats.

The film format becomes relevant, however, when it comes to how many small details are to be stored on its surface—that is a question of the total available storage capacity.

Table 1 indicates the number of pixels " constituting the height and width of the image; based on the smallest reproducible detail of 0.006mm, it gives an overview of the storage capacity of different film formats.

	Format	Width × Height (SMPTE / ISO Cameragate)	Pixels
1	S 16 mm	12.35 mm × 7.42 mm	2058 × 1237 pixels
2	S 35 mm	24.92 mm × 18.67 mm	4153 × 3112 pixels
3	65 mm	52.48 mm × 23.01 mm	8746 × 3835 pixels

1 **Super 16 mm**

2 **Super 35 mm**

3 **65 mm (70 mm print)**

Scanning

These analog image pixels must now be transferred to the digital realm. Using a Super 35 mm image as an example, the situation is as follows:
The maximum information depth is achieved with a line grid 80 lp/mm. The largest spatial frequency in the film image is therefore 1/0.012 mm (line 0.006 mm + gap 0.006 mm).

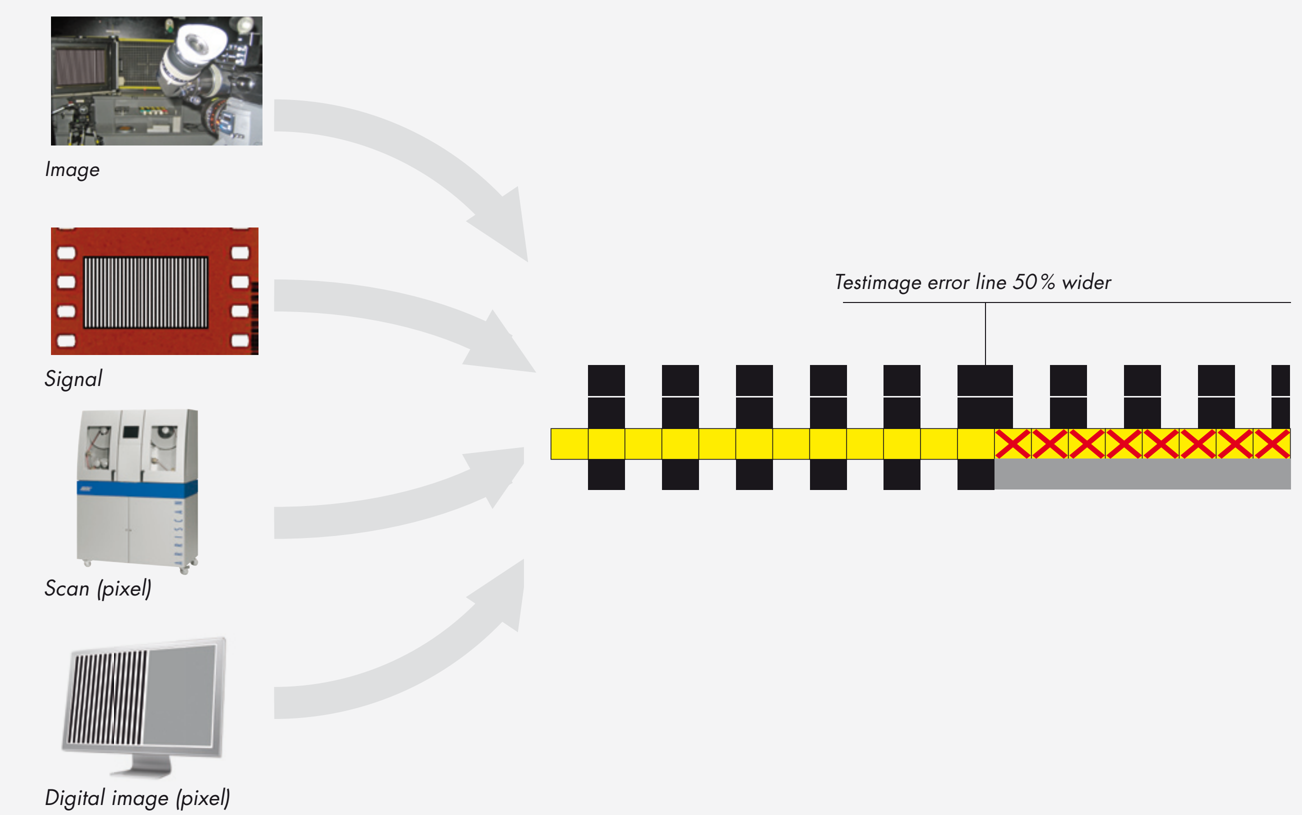
According to the scanning theorem of Nyquist and Shannon, the digital grid then has to be at least twice as fine, that is 0.012 mm/2 = 0.006 mm. Converted to the width of the Super 35 mm negative, this adds up to 24.92 mm/0.006 mm = 4153 pixels for digitalization.

Aliasing

Let us stick with our line grid as a test image and assume a small error has crept in: one of the lines is 50% wider than all the others. While the film negative reproduces the test grid unperturbedly and thereby true to the original, the regular digital grid delivers a uniform gray area, starting at the faulty line—simply because the pixels, which are marked with an "x", consist of half a black line and half a white gap, so the digital grid perceives a simple mix—which is gray.

So if the sample to be transferred consists of spatial frequencies that become increasingly higher—as in *Fig. 10*—the digital image suddenly shows lines and gaps in wrong intervals and sizes. This is a physical effect whose manifestations are also known in acoustics in the printing industry. There, the terms beating waves and moiré are in use. In digitization technology, the umbrella term for this is aliasing.

Figure 9: Principle of aliasing.



Aliasing appears whenever there are regular structures in the images that are of similar size and aligned in the same way as the scanning grid. Its different manifestations are shown in *Fig. 11*. The advantage of the "film pixel," the grain, is that they are statistically distributed and not aligned to a regular grid and so are different from frame to frame.

Figure 11 shows an area of incipient destructive interference and an area of pseudo modulation.

This describes the contrast for details which lie beyond the threshold frequency and which, in the digital image, are indicated in the wrong place (out of phase) and have the wrong size. This phenomenon does not only apply to test grids. Even the jackets of Academy Award winners may fall victim to aliasing.

Alias is a nasty artifact for still pictures, as you can see; it becomes significantly worse for motion pictures because it changes dynamically from frame to frame.

Figure 10: 6K scan of a frequency sweep.

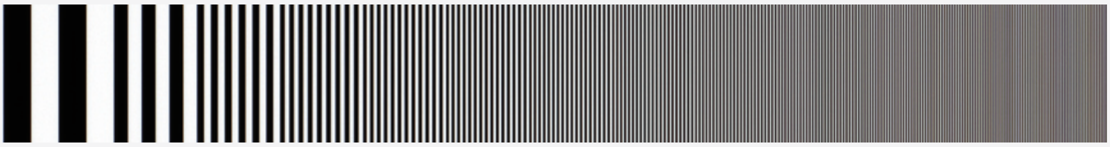


Figure 11: 3K scan with severe aliasing.

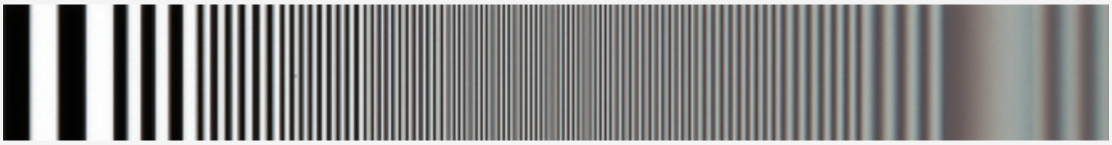


Figure 12: Digital still camera with 2 megapixels (A)

Digital still camera with 1 megapixel (B)

Impact on the MTF

It is very obvious that the aliasing effect also has an impact on the MTF. Pseudo modulation manifests itself as a renewed rise (hump) in the modulation beyond the scanning limit. ❷

Pseudo, because the resulting spatial frequencies (lines and gaps) have nothing to do with reality: instead of becoming finer, as they do in the scene, they actually become wider again in the output image *(Figs. 13 and 14).*

Figure 13: Luminance along yellow arrow.

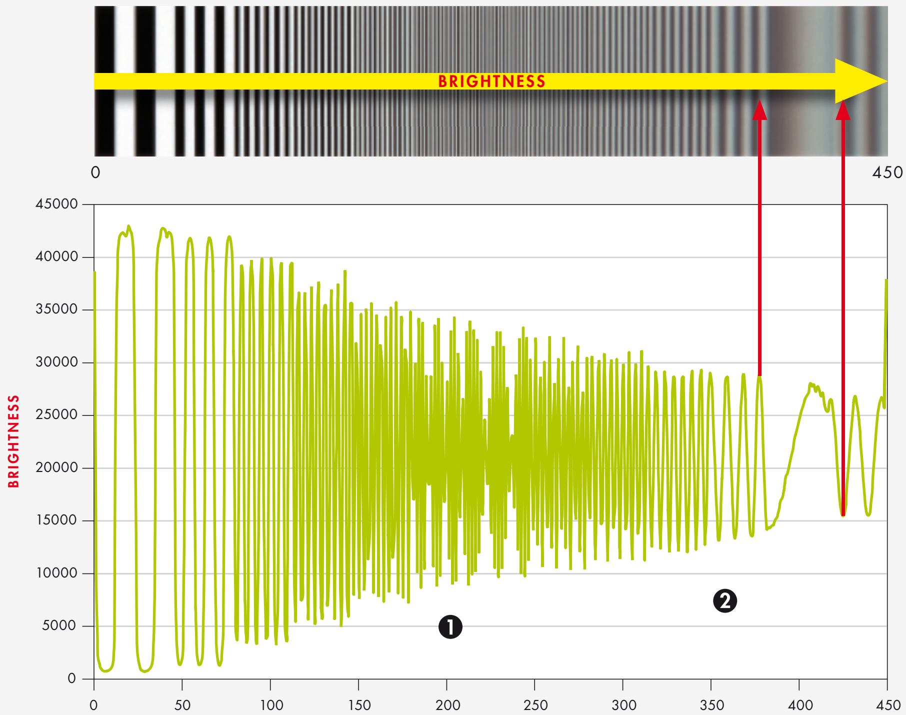
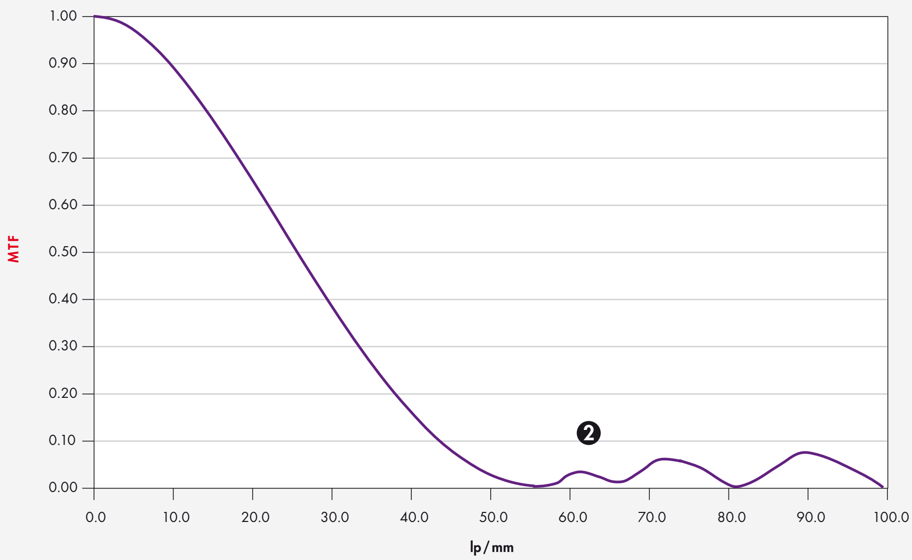


Figure 14: MTF.



Avoiding Alias

The only method to avoid aliasing is to physically prevent high spatial frequencies from reaching the scanning raster, that is, by defocusing or through a so-called optical low pass, which in principle does the same in a more controlled fashion. Unfortunately, this not only suppresses high spatial frequencies, but the contrast of coarse detail, which is important for sharpness perception, is also affected.

Another alternative is to use a higher scanning rate with more pixels, but this also has its disadvantages. Because the area of a sensor cannot become unlimitedly large, the single sensor elements have to become smaller to increase the resolution. However, the smaller the area of a sensor element becomes the less sensitive it will be. Accordingly, the acquired signal must be amplified again, which leads to higher noise and again to limited picture quality. The best solution often lies somewhere in between. A combination of a 3K sensor area (large pixels, little noise) and the use of mechanical micro-scanning to increase resolution (doubling it to 6K) is the best solution to the problem.

The currently common maximum resolution in post-production is 4K. The gained 6K data is calculated down with the help of a filter. In the process, the MTF is changed, thus:

- ❶ The response at half of the 4K scanning frequency itself is zero.
- ❷ Spatial frequencies beyond the scanning frequency are suppressed.
- ❸ The modulation at low spatial frequencies is increased.

Whereas measures ❶ and ❷ help to avoid aliasing artifacts in an an image, measure ❸ serves to increase the surface ratio under the MTF.

As discussed earlier, this improves the visual impression of sharpness. To transfer the maximum information found in the film negative into the digital realm without aliasing and with as little noise as possible, it is necessary to scan the width of the Super 35 mm negative at 6K. This conclusion can be scaled to all other formats.

Format	Width	Pixels	Scanning resolution / Digital acquisition	Final image size
S 16 mm	12.35 mm	2058 pixels	3K	2K
S 35 mm	24.92 mm	4153 pixels	6K	4K

Figure 16: 6K/4K scanner MTF.

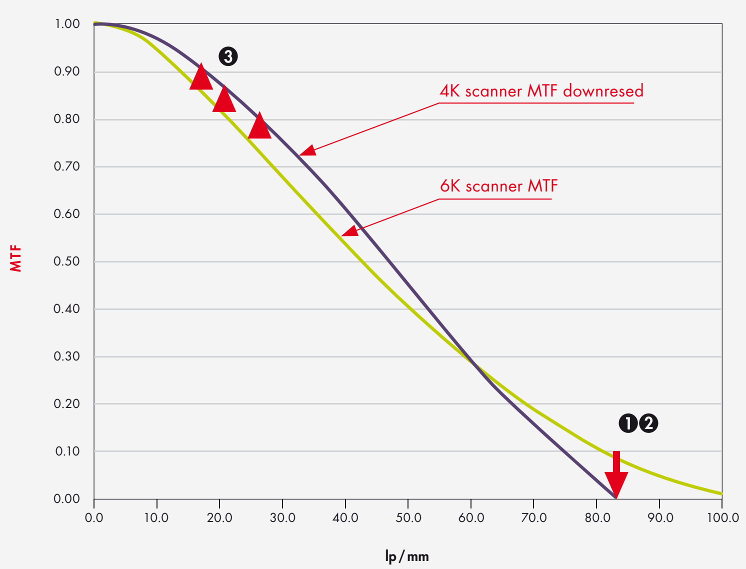
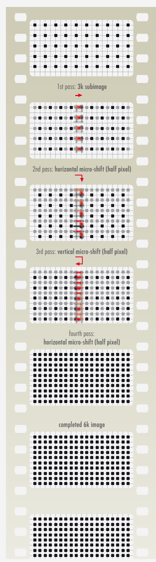


Figure 15: Microscan.



The Theory of MTF Cascading
What losses actually occur over the course of the digital, analog, or hybrid postproduction chain?
The MTF values of the individual chain elements can be multiplied to give the system MTF. With the help of two or more MTF curves one can quickly compare, without - above all - any subjectively influenced evaluation.
Also, once the MTF of the individual links of the production chain is determined, the expected result can be easily computed at every location within the chain, through simple multiplication of the MTF curves.

MTF Camera (lens) X MTF film X MTF Scanner = ?
An absolutely permissible method for a first estimate is the use of manufacturers' MTF data for multiplication. However, it must be remarked that there are usually very optimistic numbers behind this data and calculating in this way shows the best-case scenario. In this case, actual measured values are used for the calculations. Instead of using the MTF of the raw stock and multiplying it by the MTF of the camera lens, the resulting MTF of the exposed image is directly measured and multiplied by the MTF of a 4K scanner.

Figure 17: Theory of MTF cascading.

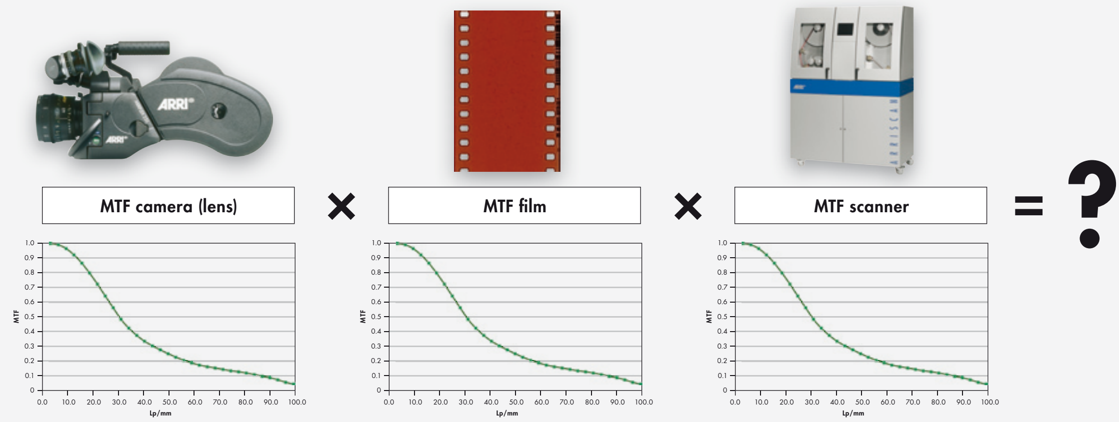


Figure 18: Resulting MTF in a 4K scan.

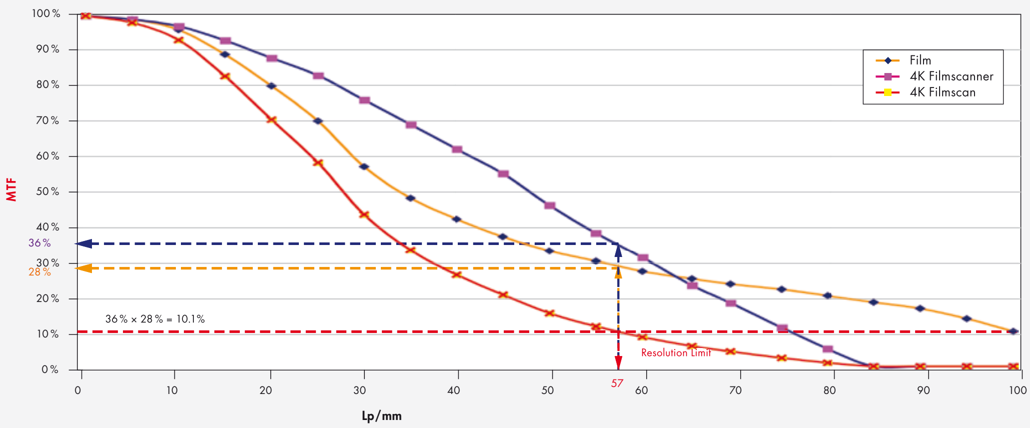


Figure 19: A true 4K production chain.

What Does the Result Show Us?

The MTF of a 35 mm negative scanned with 4K contains only little more than 56 lp/mm (the equivalent of 3K with the image width of Super 35 mm) usable modulation. The resolution limit is defined by the spatial frequency, which is still transferred with 10% modulation. This result computes from the multiplication of the modulation of the scanner and of the film material for 57 lp/mm:

MTF_4K_scanner (57 lp/mm)	×	MTF_exposed_film (57 lp/mm)	=	MTF_in_4K_scan (57 lp/mm)
36%	×	28%	=	10.08%

The same goes for a digital camera with a 4K chip. There, a low-pass filter (actually a deliberate defocusing) must take care of pushing down modulation at half of the sampling frequency (80 lp/mm) to 0, because otherwise aliasing artifacts would occur.

Ultimately, neither a 4K scan nor a (3-chip) 4K camera sensor can really transfer resolution up to 4K. This is not an easily digestible paradigm. It basically means that 4K data only contains 4K information if it was created pixel by pixel on a computer—without creating an optical image beforehand. However, this cannot be the solution, because we would then in the future have to make do with animation movies only— a scenario in which actors and their affairs could only be created on the computer. A tragic loss, not just for supermarket tabloid newspapers!

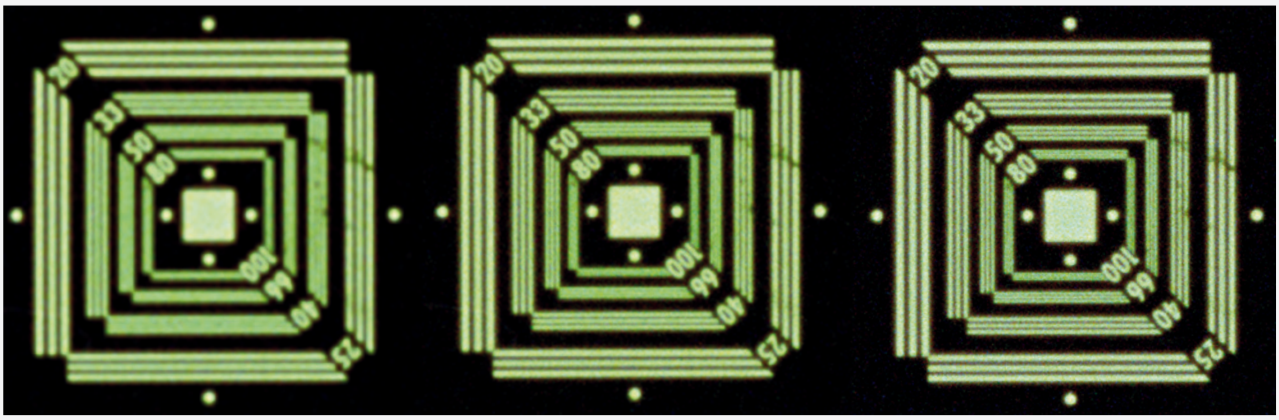
Summary Part II

4K projectors will be available in the foreseeable future, but their full quality will come into its own only when the input data provides this resolution without loss. At the moment, a correctly exposed and 6K/4K scanned 35 mm film negative is the only practical, existing acquisition medium for moving pictures that come close to this requirement. By viewing from a different angle one could say that the 4K projection technology will make it possible to see the quality of 35 mm negatives without incurring losses through analog processing laboratory technology. The limiting factor in the Digital Intermediate (DI) workflow is not the (ideal exposed) 35 mm film, but the 4K digitization. As you can see in *Fig. 17*, a gain in sharpness is still possible when switching to higher resolutions. Now imagine how much more image quality could be achieved with 65 mm (factor 2.6 more information than 35 mm)!

These statements are all based on the status quo in film technology—for an outlook on the potential of tomorrow's film characteristics read Tadaaki Tani's article.[3]

The bottomline is that 35 mm film stores enough information reserves for a digitization with 4K+.

Figure 20: 2K, 4K and 10K scan of the same 35 mm camera negative.



2K *4K* *10K*

The two most frequently asked questions regarding this subject are:

❶ ***How much quality is lost in the analog and the DI chain?***

❷ ***Is 2K resolution high enough for the digital intermediate workflow?***

The Analog Process
"An analog copy is always worse than the original." This is an often-repeated assertion. But it is true only to a certain extent in the classic postproduction process; there are in fact quality-determining parameters that, if controlled carefully enough, can ensure that the level of quality is maintained: when photometric requirements are upheld throughout the chain of image capture, creating the intermediates and printing, the desired brightness and color information can be preserved for all intents and purposes.

Clear loss of quality, however, does indeed occur where structures - that is, spatial image information - are transferred. In other words, resolution and sharpness are reduced. This occurs as a result of the rules of multiplication. *Figure 21* shows how 50 lp/mm with a native 33% modulation in the original is transferred throughout the process.

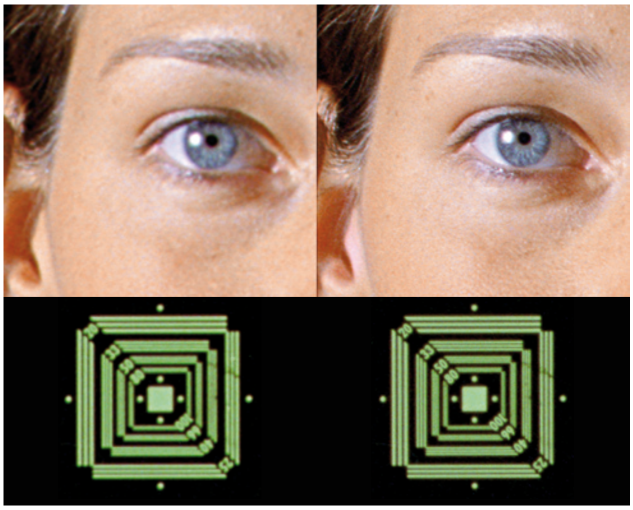
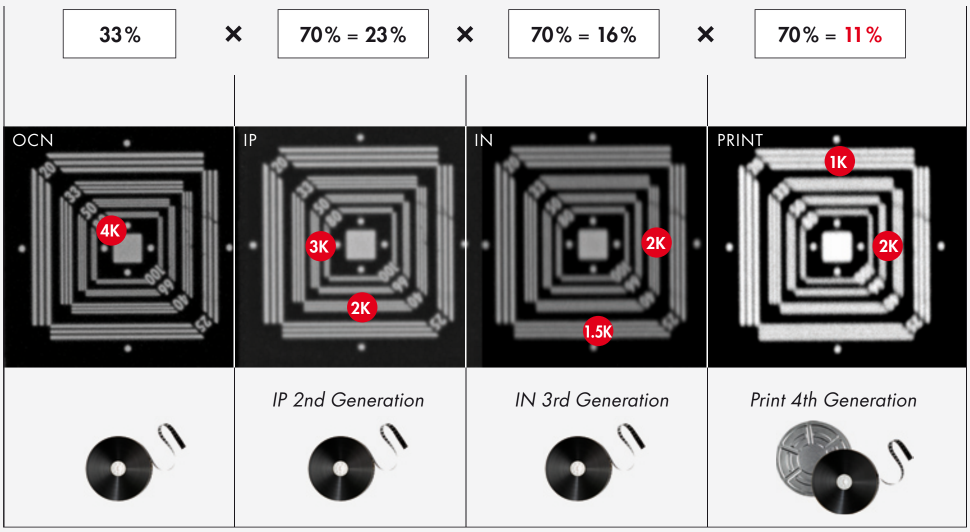


Figure 21: 2K and 4K cutout of a 2K DI and a 4K DI.

This, however, is an idealized formula, because it assumes that the modulation of the film material and the contact printing show no loss of quality. This is not the case in reality, which can be seen in the differences between the horizontal and vertical resolutions.

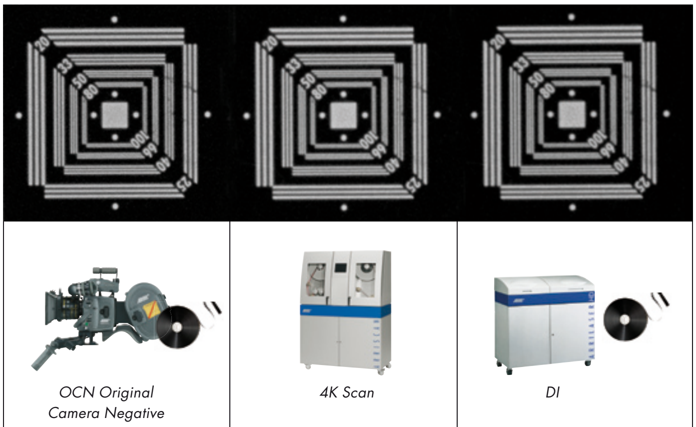
	MTF at 50lp/mm
Exposed Film Image Kodak 5205 (Camera + Film)	33%
Kodak 5242 Intermediate (Copy to IR)	70%
Kodak 5242 Intermediate (Copy to IN)	70%
Kodak Print Film 2393	70%

Figure 22: 10K scans of the image center (green).
Photochemical lab from camera neg. to screen.



Is 2K Enough for the DI Workflow?

Although the word "digital" suggests a digital reproduction with zero loss of quality, the DI process is bound by the same rules, that apply to the analog process because analog components (i.e., optical path of scanner and recorder) are incorporated. To make this clearer, let us again perform the simple multiplication of the MTF resolution limit in 4K (= 80 lp/mm). The MTF data illustrate the best filming and DI components that can currently be achieved.



	MTF at 80 lp/mm
Exposed Film Image 5205 (ARRICAM, VP, OCN)	20 %
Film Scanner MTF at 4K (ARRISCAN)	5 %
Recorded Internegative Fuji RDI (Film + ARRILASER)	20 %

20 % × 5 % = 1 % × 20 % = 0,2 %

Per definition to avoid Alias

Figure 23. DI Production chain

As the multiplication proves, inherent to the 4K DI chain, modulation cannot occur at 80 lp/mm, even though the digitally exposed internegative shows considerably more image information than can be preserved throughout the traditional analog chain.

Figure 24: MTF of a 2K filmscan.

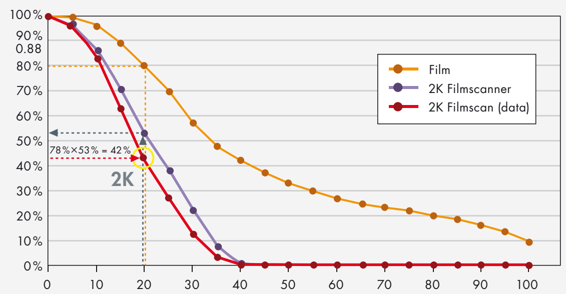
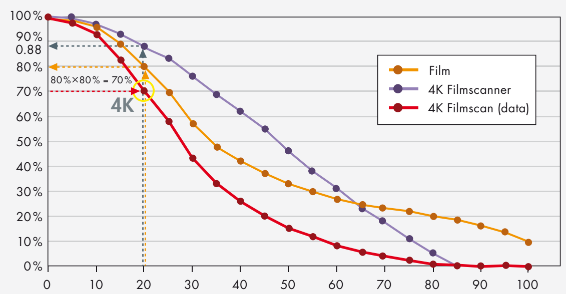


Figure 25: MTF of a 4K filmscan.



Why, then, is a 4K digital process still a good idea, even though it cannot reproduce full 4K image information? The answer is simple, according to the Heynacher Integral (the area beneath the MTF curve), the perception of sharpness depends on the modulation of coarse local frequencies. When these are transferred with a higher modulation, the image is perceived as sharp. *Figure 21* shows the results of a 2K and 4K scan.

Because a 4K scan offers not only more resolution, but also more modulation in lower local frequencies, the resulting 4K images will be perceived as being sharper. When this data is recorded out in 4K on Fuji RDI, for example, it produces the results shown in *Fig. 22.*

Summary Part III

A 4K workflow is advantageous for DI and film archiving. The MTF characteristics seen in 4K scanning will transfer coarse detail (which determines the perception of sharpness) with much more contrast than 2K scanning. It is therefore irrelevant whether or not the resolution limit of the source material is resolved. It is very important that the available frequency spectrum is transferred with as little loss as possible

PART IV: VISUAL PERCEPTION LIMITATIONS FOR LARGE SCREENS

This section discusses the most important step in the production chain—the human eye. Again, the focus is resolution, sharpness, and alias. More specifically, the focus is now on general perception at the limits of our visual systems, when viewing large images on the screen. This limitation shows how much effort one should put into digitalization of detail.

A very common rumor still circulates which alleges that one could simply forget about all the effort of 4K, because nobody can see it on the screen anyway. Let´s take a closer look to see if this is really true!

Basic parameters
Three simple concepts can be used to describe what is important for a natural and sharp image, listed here in descending order of importance:

❶ Image information is not compromised with alias artifacts.

❷ Low spatial frequencies are showing high modulation.

❸ High spatial frequencies are still visible.

This implicitly means that an alias-affected image is much more annoying than one with pure resolution. Further, it would be completely wrong to conclude that one could generate a naturally sharp image by using low resolution and just push it with a filter. Alias free images and high sharpness can be achieved only if oversampling and the right filters for downscaling to the final format have been used.

Resolution of the human eye
The fovea of the human eye (the part of the retina that is responsible for sharp central vision) includes about 140 000 sensor-cells per square millimeter. This means that if two objects are projected with a separation distance of more than 4 m on the fovea, a human with a normal visual acuity (20/20) can resolve them. On the object side, this corresponds to 0.2 mm in a distance of 1 m (or 1 minute of arc).

In practice of course, this depends on whether the viewer is concentrating only on the center of the viewing field, whether the object is moving very slowly or not at all, and whether the object has good contrast to the background.

As discussed previously, the actual resolution limit will not be used for further discussion, but rather the detail size that can be clearly seen. Allowing for some amount of tolerance, this would be around 0.3 mm at 1 m distance (= 1.03 minutes of arc). In a certain range, one can assume a linear relation between distance and the detail size.

0.3 mm in 1m distance ≈ 3 mm in 10 m distance.

This hypothesis can be easily proved. Pin the test pattern displayed in *Fig. 26* on a well-lit wall and walk away 10 m. One should be able to clearly differentiate between the lines and gaps in *Fig. 26* (3 mm). In *Fig. 27.* (2 mm) the difference is barely seen. Of course, this requires an ideal visual acuity of 20/20. Nevertheless, if you can’t resolve the pattern in *Fig. 26*, you might consider paying a visit to an ophthalmologist!

6K for 35 mm, 3K for 16 mm

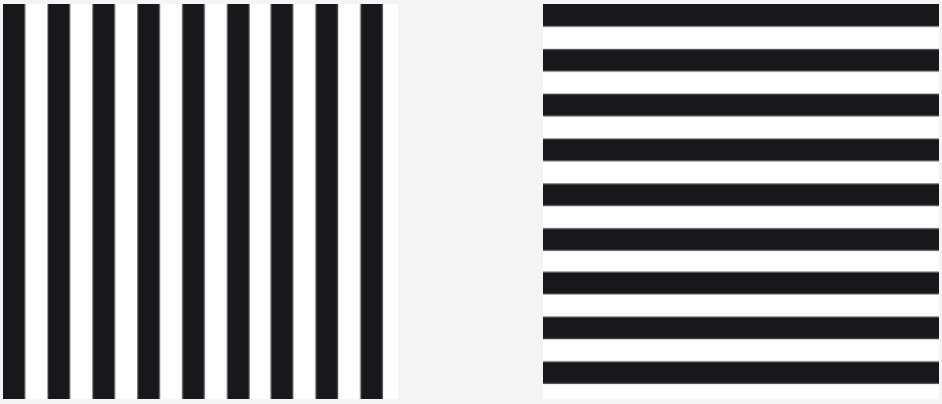


Figure 26: 3 mm

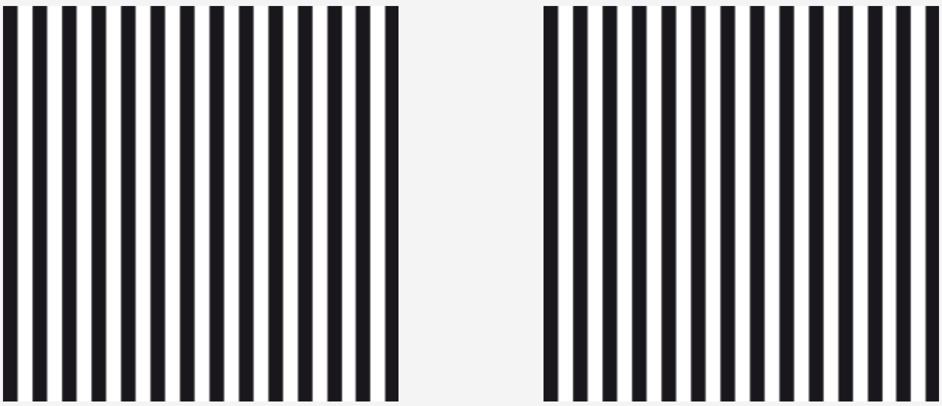


Figure 27: 2 mm

PART IV: VISUAL PERCEPTION LIMITATIONS FOR LARGE SCREENS

Large screen projection

This experiment can be transferred for the conditions in a theater. *Figure 28* shows the outline of a cinema with approximately 400 seats and a screen width of 12 m. The center row of the audience is at 10 m. An observer in this row would look at the screen with a viewing angle of 60°. Assuming that the projection in this theater is digital, the observer could easily differentiate 12 000 mm/3 mm = 4 000 pixels.

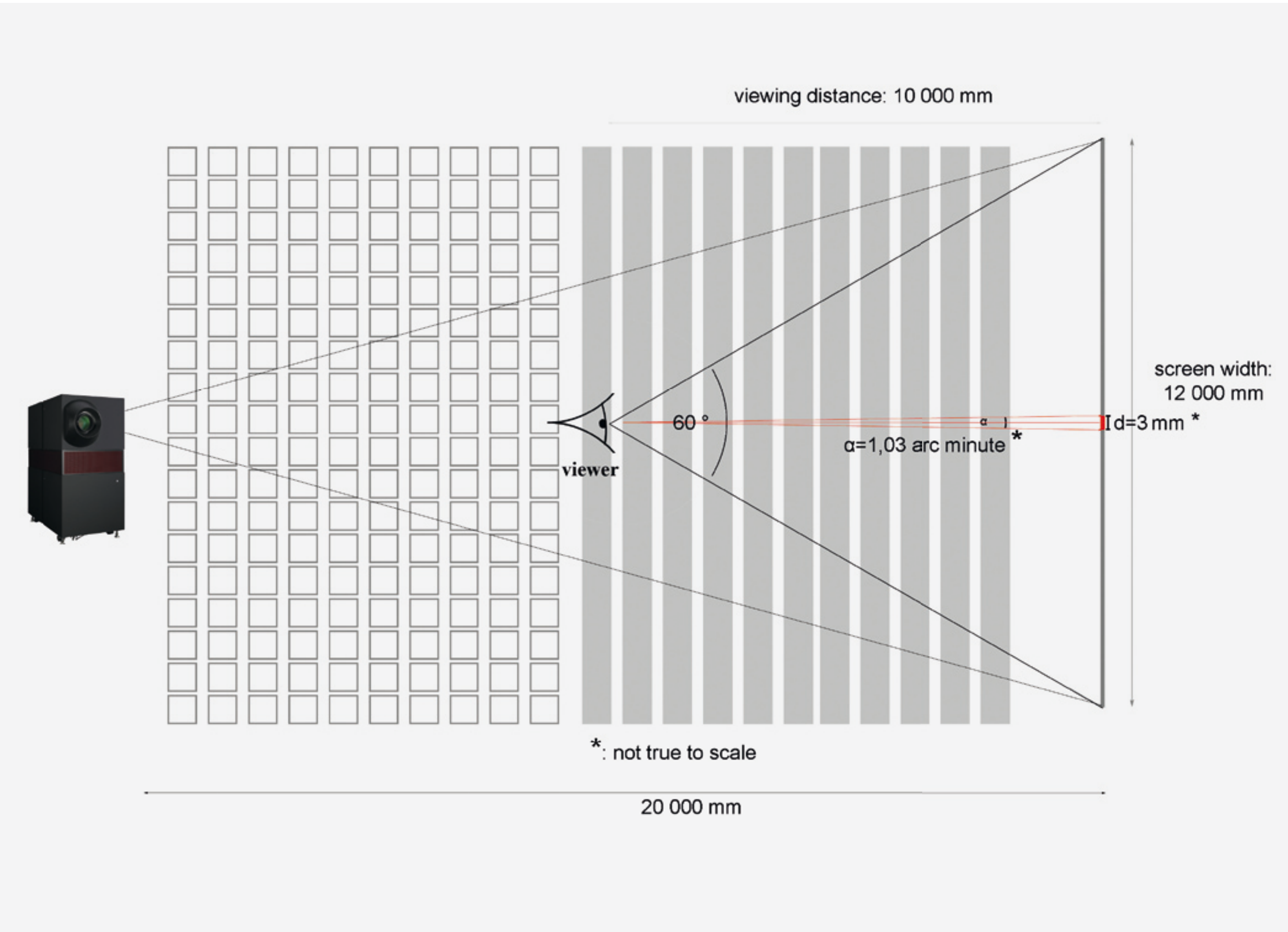
The resolution limit is not reached below a distance of 14 m. In other words, under these conditions more than 50% of the audience would see image detail up to the highest spatial frequency of the projector.

Large theaters generally have screens with widths of 25 m or more. *Figure 29* shows the amount of pixels per image width that would be needed if the resolution limit of the eye were the critical parameter for dimensioning a digital projection.

Summary Part IV

It can be concluded that the rumor is simply not true! A large portion of the audience would be able to see the 4K resolution of the projector. At the same time, the higher resolution inevitably raises the modulation of lower spatial frequencies, which in turn benefits everyone in the theater.

Figure 28: Resolution limit in the theater (how many pixels does the projector need)



Conclusion

The discussions in this paper have shown that a scientific approach is necessary to get a clear view of the real problem instead of merely counting megapixels.
This article has addressed only the static image quality factors: sharpness and resolution. As was mentioned in the introduction, there are many more parameters to consider.

For those of you who are more interested in the field of photometry and color-related image quality, I would recommend my colleague Harald Brendel's booklet "The ARRI DI Companion".

An article about the dynamic view on spatial quality parameters is also in preparation.

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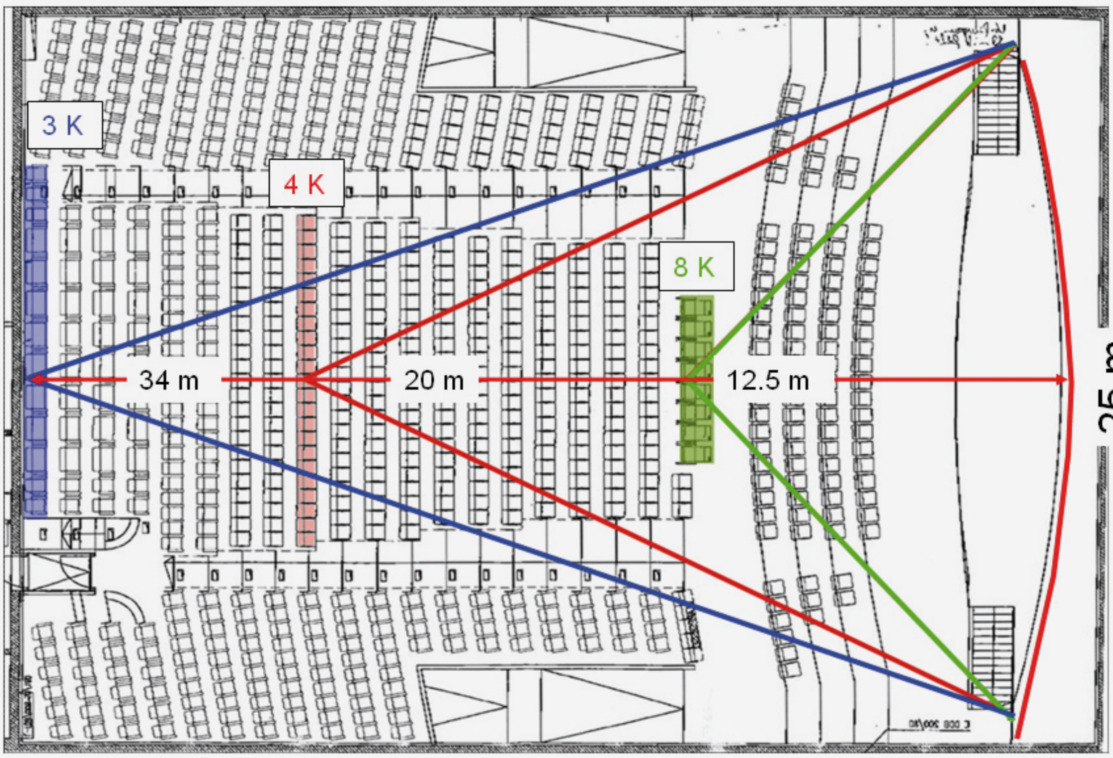
3. Tadaaki Tani, "AgX Photography: Present and Future," J. Imag. Science and Technol. 51(2):110 -116, 2007.

Figure 29: Resolution limit for large screens.



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